

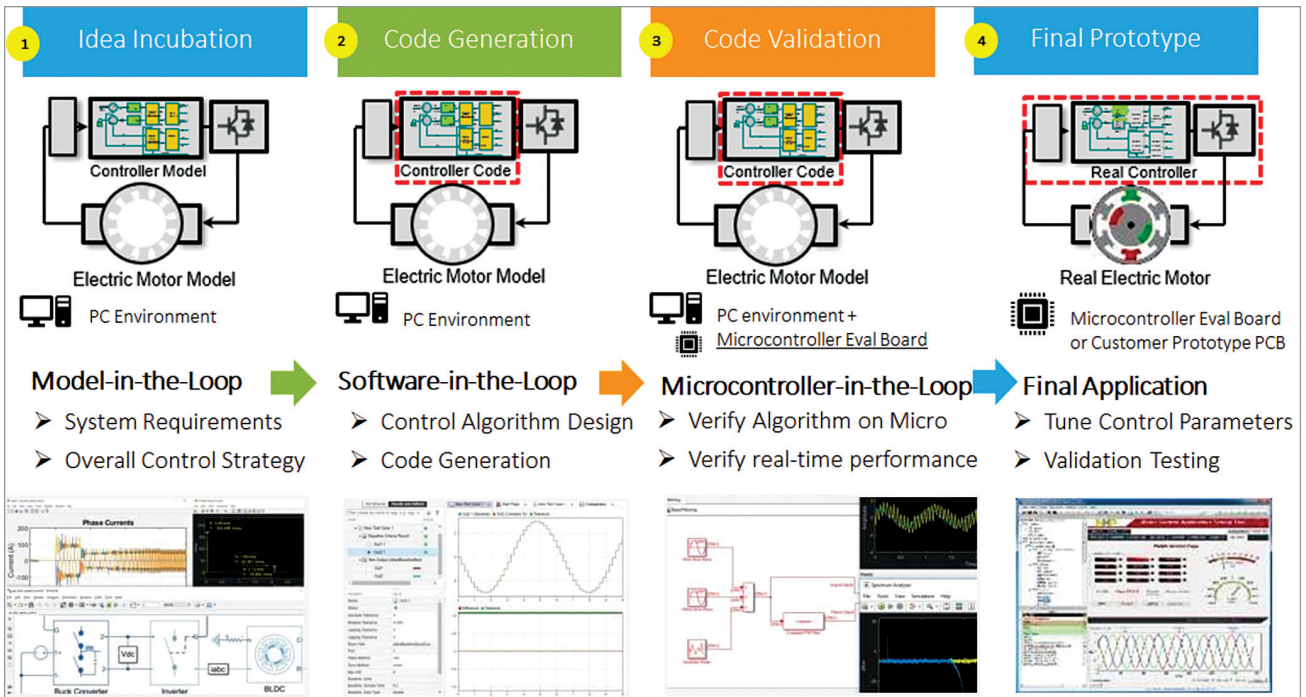


Fig. 2: Model-based design flow that enables rapid prototyping of motor control application

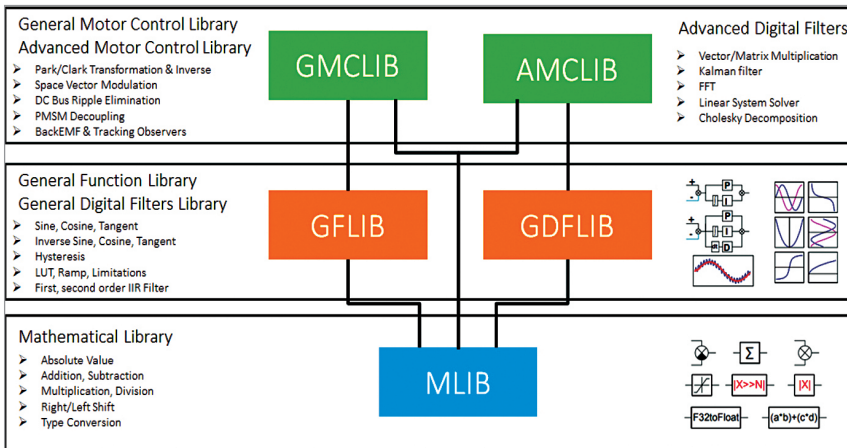


Fig. 3: Automotive math and motor control library can be crucial for accelerating time to market

motor. Some key functions of the motor controller include:

- Delivering torque in a smooth and efficient manner
- Controlling regenerative braking to extend driving range
- Offering drive modes tailored for different scenarios, such as maximum range and maximum speed

Important considerations for evaluating motor controllers include performance, efficiency, and func-

tional safety. Performance determines the vehicle driving experience, while efficiency has an impact on vehicle range and cost. Additionally, the motor controller is safety-critical since a malfunction with the system could result in a hazard. Thus, the motor controller functions are usually associated with corresponding ASIL-rated safety requirements as per ISO 26262.

If we simplify it, the motor controller fundamentally converts elec-

trical energy to mechanical energy as shown in Fig. 1.

A powerful, safety microcontroller is at the heart of the motor controller. The application-level firmware on the microcontroller implements a control process for motor control. A popular method for controlling BLDC and PMSM motors is field-oriented control (FOC), which offers the ability to deliver smooth torque with high efficiency and at high speeds—key attributes for an EV. FOC requires feedback from current and position sensors, and here, developers need to carefully consider various trade-offs associated with sensor options. Table 1 captures some key considerations for common types of sensors.

The control process sends commands to the power inverter stage of the motor controller, based on feedback signals that typically comprise inputs from current sensors and position sensors. Typically, motor controllers for electric 2W and 3W target a switching frequency of 10kHz to 20kHz, and this means the control